

Aim: To estimate the total hardness of the given sample of water by E.D.T.A

Reagent Required:

1. 0.01ml E.D.T.A. solution
2. Buffer solution of 10Ph
3. Indicator solution (Eriochrome black- T)

Theory & Reaction:

Water becomes hard due to the presence of certain dissolved salts in it like carbonates, bicarbonates, chlorides and sulphates of Calcium and Magnesium hardness prior the leathering of soap.

It is of two types

- (i) Temporary hardness: Due to the Carbonates and Bicarbonates of Calcium/Magnesium removed by boiling
- (ii) Permanent hardness: Due to the chloride and sulphate of Calcium/Magnesium can not be removed by boiling but by chemical means

Total hardness in water can be determined in laboratory by volumetric analysis of given sample of water. Water sample is titrated with E.D.T.A. solution using Buffer solution of pH 10 and indicator Eriochrome black-T. During titration, calcium present in sample reacts with E.D.T.A. and form chelated complex.

Free metal ion reacts with indicator producing complex of wine-red colour. As E.D.T.A. is a titrant to sample water, calcium and magnesium ion form chelated complex releasing free indicator. So that, at end point colour of solution changes from wine red to blue at once.

End point: At end point, colour changes from wine red to blue.

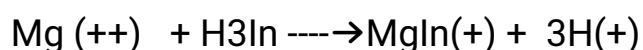
Following reaction takes place





Solution EDTA

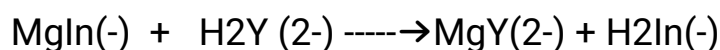
EDTA ion



Indicator blue

Wine-red colour

At end point,



Wine-red colour

Blue

Procedure:

1. Fill burette with E.D.T.A solution
2. Take 50ml of water sample with the help of pipette in a 250ml conical flask
3. Add to it 2ml of buffer solution and 2-3 drops of indicator
4. Add E.D.T.A solution from burette in small amount(drop by drop) with constant shaking of flask till the colour of solution just changes from wine red to blue with a drop of E.D.T.A. solution
5. Note the titre volume of E.D.T.A.
6. Repeat the procedure with other 50ml of sample water. Two concordant readings are observed.
7. Do the same process with distilled water in place of sample water

Note the concordant readings



Observation:

Table (i) for volume of E.D.T.A. with sample Water

S.No	Vol. of sample Water(ml)	Burette E.D.T.A. Solution		Titre Vol.(ml)	Concordant Reading
		Initial	Final		
1.	50	0	19.8	19.8	
2.	50	0	18.0	18.0	18.0
3.	50	0	18.0	18.0	

Table (ii) for volume of E.D.T.A. with distilled deionized Water

S.No	Vol.of distilled Water(ml)	Burette E.D.T.A. solution Reading		Titre Volume (ml)	Concordant reading
		Initial	Final		
1.	50	0	1.6	1.6	
2.	50	0	1.6	1.0	1.0
3.	50	0	1.6	1.0	

Calculations:

Volume of E.D.T.A. used with sample water = 18ml

Volume of E.D.T.A. used with distilled water = 1ml

Therefore, Volume of E.D.T.A. used with salts = (18-1) ml = 17ml

1ml of 0.01M E.D.T.A. = 1mg of CaCO_3

Therefore, 17ml of 0.01M E.D.T. A =

$17 \times 1 \times 1000 / \text{Volume of sample in ml (50)} = 340\text{mg/L}$

Result:

The total hardness of the given sample water is 340mg/L



Thin Layer Chromatography (TLC)

AIM

To separate and identify compounds in a mixture using thin-layer chromatography (TLC) technique.

INTRODUCTION

Thin-layer chromatography (TLC) is a rapid, simple, and inexpensive technique for separating mixtures into individual components. It is widely used for qualitative analysis of organic compounds and monitoring the progress of reactions.

MATERIALS NEEDED

- TLC plates (silica gel or alumina coated)
- Solvent mixture (e.g., ethyl acetate, hexane, or other suitable solvent system)
- Capillary tubes or micropipettes for spotting samples
- Developing chamber (jar or beaker with a lid)
- Filter paper
- UV lamp or iodine chamber for visualizing spots
- Ruler, pencil, and tweezers
- Samples (e.g., dye mixtures, plant extracts)
- Gloves, safety goggles, and lab coat

PROCEDURE

1. Preparation of the TLC Plate:

- Handle the TLC plate by the edges to avoid contamination.
- Draw a light pencil line about 1-2 cm from the bottom of the plate (baseline).
- Mark several points on this line where you will spot the samples.

2. Spotting the Samples:

- Use a capillary tube or micropipette to transfer a small amount of each sample onto the marked spots on the baseline.
- Ensure the spots are small and concentrated by lightly tapping the capillary on the plate.
- Let the spots dry before proceeding.

3. Preparation of the Developing Chamber:

- Pour a small amount of solvent mixture (mobile phase) into the chamber to a depth of about 0.5 cm.
- Line the inside of the chamber with a piece of filter paper to saturate the atmosphere with solvent vapors.
- Close the chamber and allow it to equilibrate for a few minutes.

4. Developing the Plate:

- Carefully place the spotted TLC plate into the developing chamber, ensuring that the solvent level is below the baseline.
- Close the chamber and allow the solvent to rise up the plate by capillary action.
- Once the solvent front is near the top of the plate (about 1 cm from the edge), remove the plate and quickly mark the solvent front with a pencil.

5. Visualization of Spots:

- Allow the plate to dry.
- Visualize the spots under a UV lamp or in an iodine chamber.
- Circle the observed spots with a pencil for further analysis.

6. CALCULATING THE R_f VALUES:

- Measure the distance traveled by each spot (from the baseline to the center of the spot).
- Measure the distance traveled by the solvent front.
- Calculate the retention factor (R_f) for each spot using the formula:

$$R_f =$$

SAFETY PRECAUTIONS:

- Wear gloves, goggles, and a lab coat at all times.
- Work in a well-ventilated area or fume hood when handling solvents.
- Dispose of solvent waste properly according to your lab's safety protocols

OBSERVATION TABLE

Sample no.	Sample Name	Distance travelled by solute(cm)	Distance travelled by solvent(cm)	R _f Values	Spot color(under uv\iodine)	No. of Spots
1.	Methylene Blue	2.5	5.0	0.50	Blue (under uv)	1
2.	Eosin	3.2	5.0	0.64	Pink(under uv)	1
3.	Mixture (Methylene Blue,	2.5 MB\3.2 Eosin	5.0	0.50\0.64	Blue\Pink	2

Conclusion:

This TLC experiment demonstrates how different compounds (methylene blue and eosin) can be separated based on their retention factors (R_f values) and visualized using UV light. The mixture was successfully resolved into its two components.

EXPERIMENT -2

Removal of hardness in water using Ion exchange column

AIM: To remove hardness from water using an ion exchange column and to determine the efficiency of the process.

THEORY

Hardness in water is caused primarily by the presence of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. Ion exchange is a common method for softening water, where these ions are replaced with sodium (Na^+) ions.

APPARATUS

1. Ion exchange column
2. Water sample with known hardness
3. Regenerant solution (e.g., sodium chloride solution)
4. Measuring cylinders
5. Beakers
6. pH meter or test strips
7. Hardness testing kit or titration setup
8. Flow control valves

CHEMICALS

1. Sodium chloride (NaCl)
2. Water sample

PROCEDURE

1. **Preparation of Ion Exchange Column:**
 - Fill the ion exchange column with the ion exchange resin.
 - Wash the resin with deionized water to remove any impurities.
2. **Softening Process:**
 - Pass the hard water sample through the ion exchange column at a controlled flow rate.
 - Collect the effluent (treated water) in a beaker.
3. **Regeneration Process:**
 - After treating a certain volume of hard water, regenerate the resin by passing a regenerant solution (e.g., 10% NaCl solution) through the column.
 - Rinse the column with deionized water to remove excess regenerant.
4. **Testing and Observation:**
 - Test the hardness of the influent (before treatment) and effluent (after treatment) water samples using a hardness testing kit or titration method.

OBSERVATION TABLE

Sample	Volume of Water treated (ml)	Initial Hardness (ppm)	Final Hardness (ppm)	Volume of regenerant used (ml)
1.	500	300	20	100
2.	500	300	25	100
3.	500	300	15	100

1. The initial hardness of the water sample was consistently 300 ppm.
2. The final hardness of the effluent water varied between 15 to 25 ppm.
3. The ion exchange column was regenerated after treating every 500 mL of hard water using 100 ml of regenerant solution.
4. The efficiency of the ion exchange column in removing hardness can be calculated by comparing the initial and final hardness values.

CALCULATION

Efficiency Calculation

$$\text{Efficiency (\%)} = \left(\frac{\text{Initial Hardness} - \text{Final Hardness}}{\text{Initial Hardness}} \right) \times 100$$

For sample 1:

$$\begin{aligned} \text{Efficiency (\%)} &= \left(\frac{300 - 20}{300} \right) \times 100 \\ &= 93.33\% \end{aligned}$$

$$\text{Average Efficiency (\%)} = \frac{\Sigma \text{Efficiency}}{\text{Number of sample}}$$

Using the sample data:

$$\begin{aligned} \text{Average Efficiency (\%)} &= \frac{93.33 + 91.67 + 95}{3} \\ &= 93.33\% \end{aligned}$$

RESULT

The ion exchange column effectively reduced the hardness of water sample from 300 ppm to an average of 20 ppm, demonstrating an average efficiency of 93.33%.

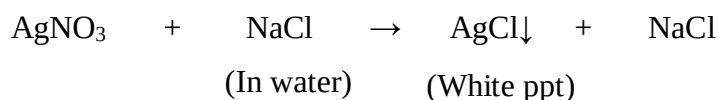
EXPERIMENT – 7

Chloride content of water by Mohr's Method

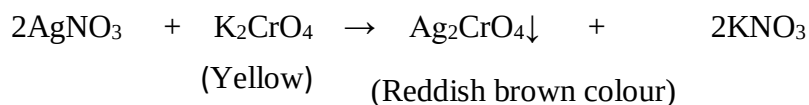
AIM – TO determine the amount of chloride present in 100 ml of the water sample.

PRINCIPLE- The water contains ions in the form of NaCl, KCl, CaCl₂ and MgCl₂. The maximum advisable limit of chloride ion in water is 250 ppm and above which the water is desirable for drinking purpose. The amount of chloride ion in water can be determined by Argentometric or Mohr's method.

In this method, the chloride ion solution is directly titrated against silver nitrate solution (AgNO₃) using potassium chromate as indicator.



At the end point, when all the chloride ions are removed, the yellow color of chromate changes to reddish brown due to the formation of silver chromate (Ag₂CrO₄).



Titration –I

Standardization of silver nitrate.

Titration –II

Determination of chloride ions in water.

Titration –I: Standardization of silver nitrate

Burette solution: Silver nitrate

Pipette solution: Standard sodium chloride

Indicator: Potassium chromate

End point: Reddish brown

Titration –II: Determination of chloride ion.

Burette solution: Standard silver nitrate

Pipette solution: Water sample

Indicator: Potassium chromate

End point: Yellow color to reddish brown

Calculation:

For the determination of chloride content in water

Titration – I: standardization of silver nitrate (standard NaCl vs. AgNO₃)

Indicator: K₂CrO₄

Volume of NaCl (V₁) = 20ml

Strength of NaCl (N₁) = 0.01

Volume of AgNO₃ (N₂) =?

SN	Volume of NaCl (in ml)	Burette reading in ml		Volume of AgNO ₃ Consumed (ml)	Concordant volume (ml)
		Initial	Final		
1.	20	0	19.5	19.5	19.5
2.	20	0	19.5	19.5	

According to law of volumetric analysis,

$$V_1N_1 = V_2N_2$$

$$N_2 = \frac{V_1N_1}{V_2}$$

$$N_2 = \frac{20 \times 0.01}{19.5}$$

$$= 0.0102N$$

Strength of Silver Nitrate = 0.0102N

Titration –II Determination of chloride ion in water (Standard AgNO₃ VS. Water sample)

Indicator - K₂CrO₄

SN.	Volume of water sample V ₂ (ml)	Burette Reading (ml)		Volume of AgNO ₃ Consumed	Concordant Volume(ml)
		Initial	Final		
1.	20	0	9.5	9.5	9.5
2.	20	0	9.5	9.5	

Volume of AgNO₃ (V₁) = 9.5 ml

Strength of AgNO₃ (N₁) = 0.0102N

Volume of water sample (V₂) = 20ml

Strength of water sample (N₁) = ?

According to the law of volumetric analysis

$$V_1N_1 = V_2N_2$$

$$N_2 = \frac{V_1N_1}{V_2}$$

$$= \frac{9.5 \times 0.0102}{20} = 0.0048$$

Strength of water sample = 0.0048N

Amount of chloride ion present in 1 liter (1000ml) Of the water sample

= Strength of water sample x Equivalent weight of chloride

$$= 0.0048 \times 35.46 \times 1000$$

$$= 170 \text{ ppm}$$

Amount of Chloride ion present in 1000ml of the water sample

$$= 170 \times \frac{100}{1000} \text{ ppm}$$

= 17 ppm

Result

(i) Amount of chloride ion in whole of the given water sample = 170ppm

(ii) Amount of chloride ion present in 100ml of water sample = 17 ppm.

EXPERIMENT – 4

Determination of P^H by P^H Meter

AIM – To determine P^H of given solution by P^H - meter.

APPARATUS REQUIRED

1. Conical flask
2. Conical beaker
3. P^H – meter

CHEMICAL REQUIRED

1. Given solution (A)
2. Given solution (B)

THEORY

P^H may be defined as the negative logarithm of $[H^+]$ concentration.

Or

It is the negative logarithm (base 10) of the hydronium ion concentration in mol L^{-1} .

$$P^H = -\log [H^+]$$

Or

$$P^H = -\log [H_3O^+]$$

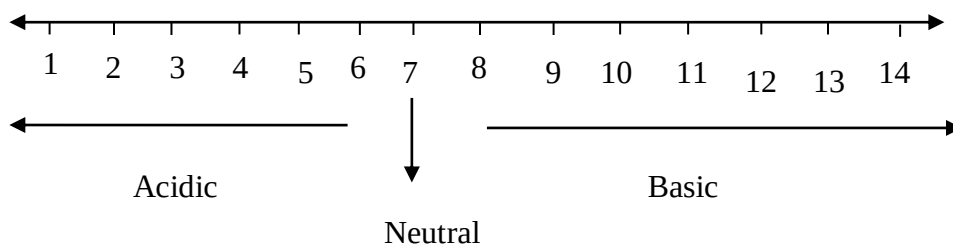
For neutral solution

$$P^H = -\log 10^{-7} = 7$$

For an acidic solution, $P^H < 7$

For a basic solution, $P^H > 7$

P^H SCALE



PROCEDURE

Given solution was taken in a conical flask .Then an electrode was dipped into the given solution .After that, reading of the P^H was noted.

OBSERVATION TABLE

Given solution	PH Reading	Conclusion
Solution (A)	5.6	Acidic
Solution (B)	7.8	Basic

RESULT

Solution (A) - Acidic

Solution (B) - Basic

Aim: - Preparation of Phenol-Formaldehyde resin (Bakelite).

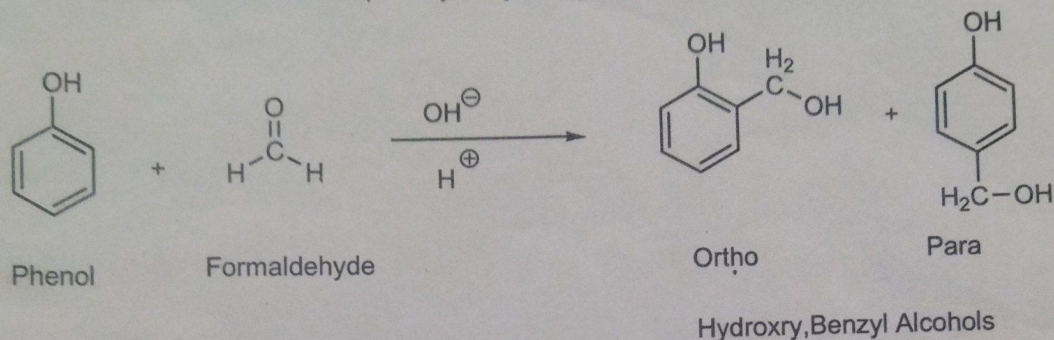
Apparatus: - Beaker, Conical flask, glass rod, measuring cylinder, fractional weight box etc.

Chemicals: - Glacial acetic acid, 40 % formaldehyde solution, phenol, conc. HCl, Dist. water etc.

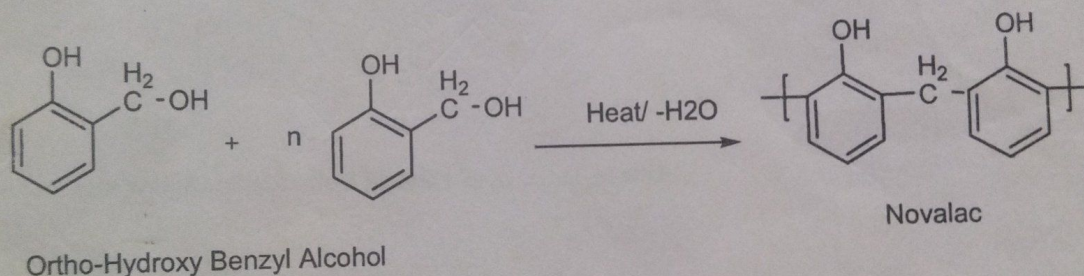
Theory: - Phenol resins are condensation polymerization product of phenolic derivative with aldehyde (like formaldehyde, furfural) It is prepared by condensing phenol with formaldehyde in presence of acid or alkaline catalyst.

Reaction:-

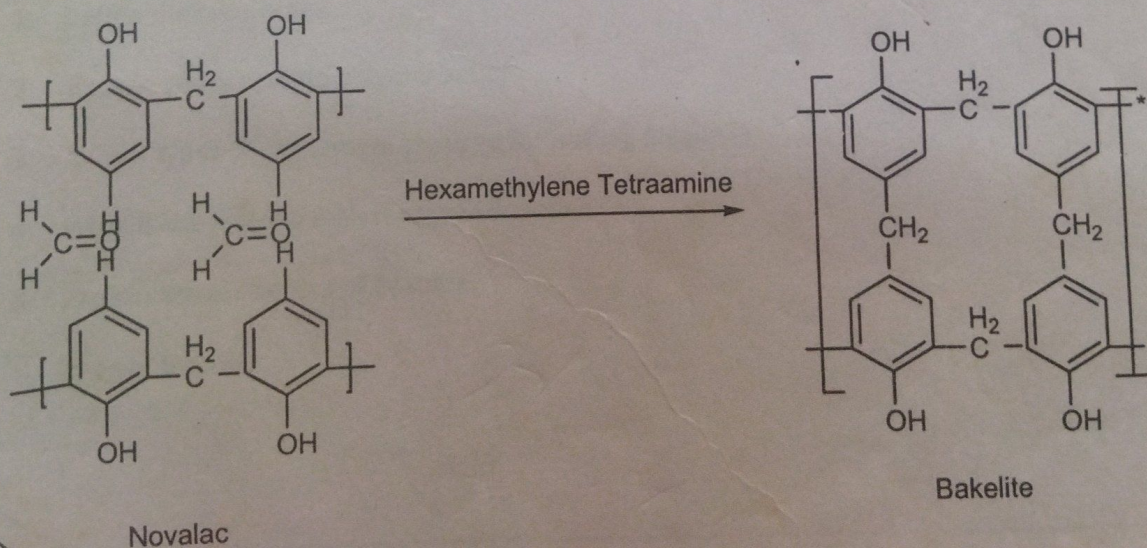
Step I:- Formation of ortho and para hydroxy benzyl alcohols from phenol and formaldehyde:-



Step II:- Formation of Novalac from ortho-hydroxy, benzyl alcohol



Step III:- Formation of Bakelite from Novalac:-



Procedure:-

- 1) Place 5 ml of glacial acetic acid and 2.5 ml of formaldehyde solution in 500 ml beaker.
- 2) Add 2 gr of phenol and 1 ml of conc. HCl solution in it.
- 3) Heat the solution slowly with constant stirring for 5 minutes.
- 4) A large mass of pink Plastic is formed.
- 5) The residue obtained is washed several times with distilled water.
- 6) Dry the product and calculate the yield accurately.

Result: - The weight of obtained Bakelite is grams.

Question-

1. Define the term resin.
2. Explain condensation polymerization.
3. Which types of monomers are used for making Bakelite?
4. What is the role of catalyst in the reaction?
5. Explain the properties of Bakelite.
6. Uses of Bakelite.

Name of the experiment: Qualitative Analysis of unknown salt Mixture No. 'B'.

A. Dry Test for Basic Radicals

1. Physical Characteristics:

- (i) Colour - Green.
- (ii) State - Amorphous.
- (iii) Solubility in water - Insoluble.

2. Action of heat:

Obs.	Inf.
(i) Colour changed from green to black.	Cu may be.
(ii) Brown gas evolved, turned FeSO_4 solution brown.	oxides of nitrogen from heavy metals nitrate.
(iii) Cracking sound.	oxidising agents, usually NO_3^- .

3. Charcoal Test in R.F.:

Obs.	Inf.
(i) Gray metallic bead melted paper, yellow incrustation.	Pb may be
(ii) Red metallic scales.	Cu may be
In O.F. Infusible incandescent mass. not found.	Ca, Sr, Ba, Al, Zn, Mg, PO_4^{3-} absent.

4. Flame Test :

Obs.	Inf.
Green with blue centre	Cu may be.

5. Borax Bead Test :

Obs.	Inf.
(i) In O.F. : Green when hot, blue when cold.	Cu may be
(ii) In R.F. : Colourless when hot, opaque when cold.	Cu may be

B. WET TEST FOR BASIC RADICALS

(1) Solution Preparation : The mixture was soluble in dil. HNO_3 accompanied by effervescence.

2. Group Separation : Dilute HCl was added to cold solution. White Precipitate.
Precipitate was washed and filtered.

Group I	Filtrate was heated and then H_2S gas was passed, filtered and washed.			
Present	Black Precipitate	H_2S was boiled off from the filtrate + 2 ml conc. HNO_3 and boiled. $NH_4Cl(s)$ was added, boiled and excess of NH_4OH solution was added.		
Group II	Present	No Precipitate	H_2S gas was passed in the filtrate.	
		III A absent	No Precipitate	H_2S was boiled off from the filtrate + NH_4OH + $(NH_4)_2CO_3$ solution and allowed to stand.
		III B absent	No Precipitate	The filtrate was treated with sodium hydrogen phosphate and NH_4OH .
			Group IV absent	No Precipitate $\rightarrow Mg^{2+}$ absent.

(3) (i) Analysis of Group I: The Precipitate was washed first with dil HCl and then with cold water. The Precipitate was boiled and then filtered.

No Precipitate. Hg^{2+} and Ag^+ absent.	The hot filtrate was divided into two parts. Part I: $\rightarrow$ Allowed to cool - white needle shaped crystals, Pb^{2+} Present. Part II: $\rightarrow$ Acidified with dil. $CH_3COOH + K_2CrO_4$ solution - Yellow precipitate $\rightarrow Pd^{2+}$ confirmed.
---	---

(ii) Analysis of Group II: The precipitate was warmed with yellow ammonium sulphide solution for 5 minutes and then filtered.

Black precipitate, Group II Present.	Filtrate was acidified with conc. HCl and warmed gently. Fine white ppt. of sulphur $\rightarrow$ Group II B absent.
---	--

(iii) Analysis of Group II A: The Precipitate was heated with dil. HNO_3 .

No ppt. Hg^{2+} absent.	An excess of NH_4OH solution was added to the remaining solution.	
	NO ppt. Bi^{3+} absent	Blue solution was obtained — Cu^{2+} present. Cu was confirmed by adding $\text{K}_4\text{Fe}(\text{CN})_6$ solution and dil. acetic acid, brown ppt. No yellow ppt. — Cd^{2+} absent.

C. DRY TESTS FOR ACID RADICALS

Exp.	Obs.	Inf.
1. Action of dil. HCl on salt mixture.	Brisk effervescence in cold, colourless, odourless gas evolved, lime water turned milky.	CO_2 from CO_3^{2-} .
2. Action of conc. H_2SO_4 on salt mixture.	Yellowish brown gas with nitrous smell evolved. On adding Cu turning, colour became more intense. Brown gas turned FeSO_4 solution black.	Nitrate present because no brown gas was evolved with dil. acid.

D. WET TESTS FOR ACID RADICALS :

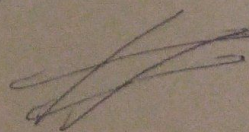
Exp.	Obs.	Inf.
(i) 1-2 ml extract + dil. HCl. CO_2 boiled off and then BaCl_2 solution was added.	No white precipitate	SO_4^{2-} absent.
(ii) 1-2 ml extract + dil. HNO_3 . CO_2 boiled off and AgNO_3 solution was added.	No white precipitate	Halides, Sulphides absent.

Ring test was positive - NO_3^- radical confirmed.

Conclusion : Hence the given salt mixture contains :

Basic Radicals - Pb^{2+} and Cu^{2+}

Acid Radicals - NO_3^- and CO_3^{2-}





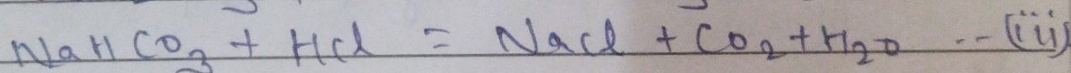
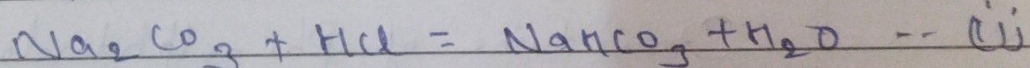
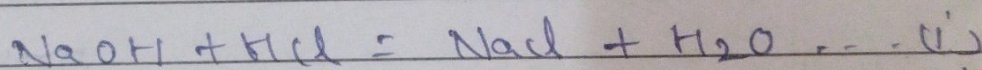
EXPERIMENT No. - 1

Aim - Determination of NaOH and Na_2CO_3 in a given alkali mixture.

Materials Required:

1. Bottle A, = Alkali mixture
2. Bottle B = Standard HCl
3. Phenolphthalein (HPh) indicator
4. Methyl orange indicator

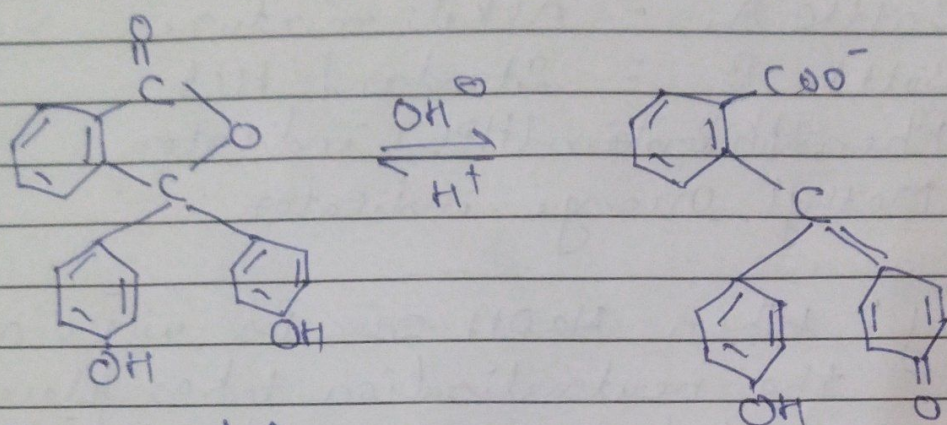
Theory: When NaOH reacts with an acid the neutralisation takes place in one stage, but in case of Na_2CO_3 the neutralisation takes place in two stages.



Phenolphthalein acts as an indicator for reactions (i) & (ii) successfully, but as soon as the reaction (iii) starts, the liberation of CO_2 discharges its colour and it ceases to act as indicator.

If methyl orange is added at this stage it may act successfully in indicating the end point. Thus, the

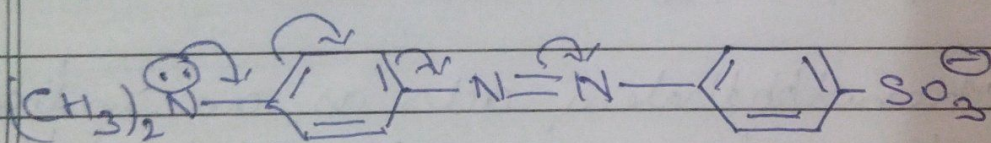
volume of acid used with phenolphthalein as indicator will correspond to whole of the NaOH and half of the Na_2CO_3 .



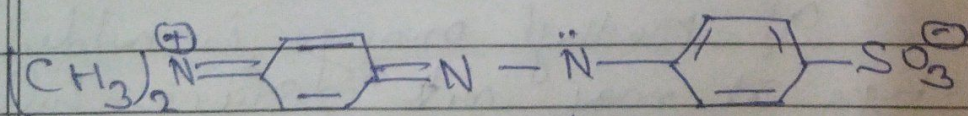
Benzenoid form
(Colourless in Acidic medium)

Quinonoid form
(Pink color in Basic medium)

Structure of Phenolphthalein Indicator.



Benzenoid form
(Yellow color in Basic medium)
⇌



Quinoid form
(Orange Red color in Acidic medium)

Structure of Methyl orange Indicator.



SANDIP
UNIVERSITY

SANDIP UNIVERSITY

At Village : Sijoul, Post : Mailam, Dist.: Madhubani, Bihar - 847235

Date: _____ Page _____

3

Procedure: 10ml of bench NaOH and 10ml of bench Na_2CO_3 solution were taken in a 250ml of measuring flask. The solution was diluted upto the mark with constant shaking. The solution was allowed to stand for sometime. 25ml of the mixture was taken into a conical flask and titrated against $\text{N}/10$ HCl using Phenolphthalein indicator until the pink colour was just discharged. The burette reading was noted. Then 1-2 drops of methyl orange indicator was added to it. It was now titrated against the $\text{N}/10$ acid solution. Three concurrent readings were noted.

Burette Reading in ml.

No. of obs	Phenolphthalein indicator				Methyl orange indicator			
	Initial	Final	Diff	Const.	Ini.	Final	Diff	Const.
1.	0.0	29.8	29.8		29.8	39.8	9.9	
2.	5.0	34.8	29.8	29.8	39.8	49.7	9.8	9.9
3.	10	39.8	29.8		39.8	49.7	9.9	

Original volume of the solution = 20ml
 volume of the diluted mixture solution = 250ml
Calculation

